

SC 107 Calculus

Course Placement: Calculus is a core course for first year BTech ICT program.

Course format: It is **3 hours lecture** every week and **1** tutorial per week.

Credit Structure: 3-1-0-4 (L-T-P-C)

Course content: Functions of single variable-Mean value theorems and Taylor's theorem, Fundamental theorem of integral calculus, Functions of several variables-Continuity, Differentiability, maxima, minima, Multiple integrals, Complex Variables-Differentiability and analyticity, definite integrals (contour integrals-line integrals), Cauchy integral theorem and formula, Taylor and Laurent series, zeroes, singularities and residues, Ordinary differential equations-ODE of first order, linear ODE of second and higher order with constant and non-constant coefficients.

Suggested Textbook/references:

1. Thomas's Calculus (Textbook)
2. Introduction to Real Analysis, Robert G. Bartle Donald R. Sherbert
3. A basic course in real analysis, Ajit Kumar and S. Kumaresan
4. Differential equations, S. L. Ross
5. A first course in Complex Analysis with applications, D. G. Zill and P. D. Shanahan

Assessment method: Two in-sem examinations (30% each) and a final examination (40%).

Course Outcomes:

The students after completing the course will get a basic overview of Calculus and its applications. They will get an insight how it is used in various applications.

Through this course students can develop ability to

- ☐ Apply the knowledge of mathematics, science, engineering fundamentals, and engg. specialization to the solution of complex engineering problems (**P1**)
- ☐ Understand and create mathematical arguments for solving problems (**P2**)
- ☐ Understand mathematical structures such as functions, variable, integrations and learn their uses (**P12**)
- ☐ Develop skills towards mathematical modeling and analysis of engineering problems (**P3, P4**)

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X								X

Detailed Course Content

Topic	Content	No. of Lectures
Calculus of single variable	Countable, uncountable sets, sequence, series, limit, continuity, differentiability, application of differentiability, Rolle's theorem, Mean value Theorem, Riemann integral, Fundamental Theorem of Calculus	15
Multivariable calculus	Functions of several variables, Limit, Continuity, Differentiability, Partial derivative, Double integral, Triple integral	8
Complex Analysis	Complex functions, limit, continuity, C-R equation, Analytic function, Zeros of functions, singularity, Contour integral, Cauchy integral theorem, Residue	13
Ordinary differential equation	ODE of first order, Existence and uniqueness theorem, Exact differential equation, Integrating factor, Separation of variables, Homogeneous equation, 1 st order linear ODE, Wronskian, Linear ODE with constant coefficients.	6